ch24 Electromagnetic Induction

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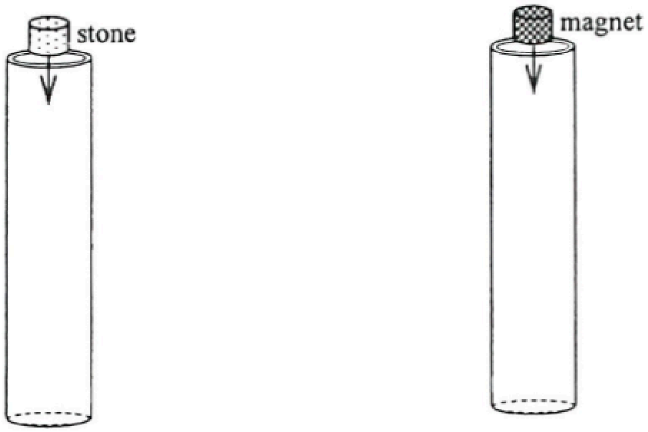
2018 Q29

29.

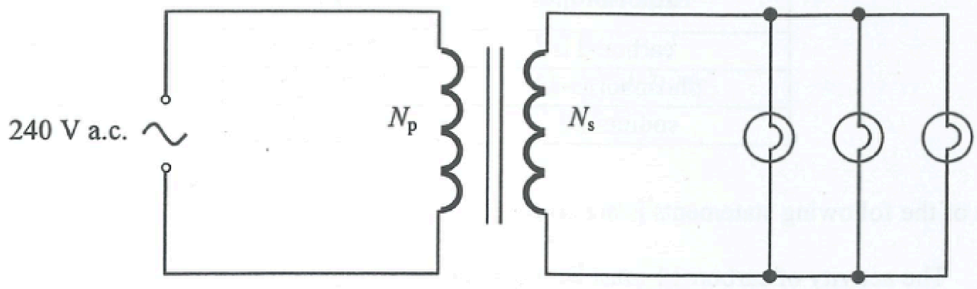


A stone and a strong magnet of the same size and shape are released from rest into a hollow aluminium tubing respectively. Which of the following is correct ? Neglect air resistance.

	drops slower	reason
A.	stone	the stone is more massive
B.	magnet	the stone is more massive
C.	stone	the magnet induces eddy current in the aluminium tubing
D.	magnet	the magnet induces eddy current in the aluminium tubing

2019 Q29

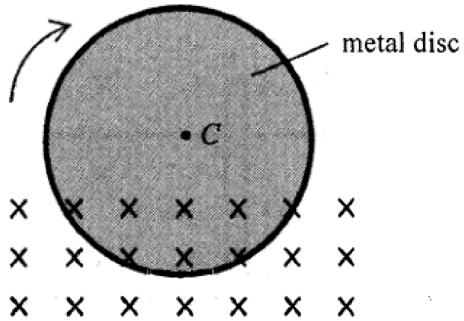
*29. In the circuit below, each light bulb works at rated power '12 V 24 W'. What should be the turns ratio (N_p:N_s) of the transformer ?



A. 40 : 1
B. 30 : 1
C. 20 : 1
D. 10 : 1

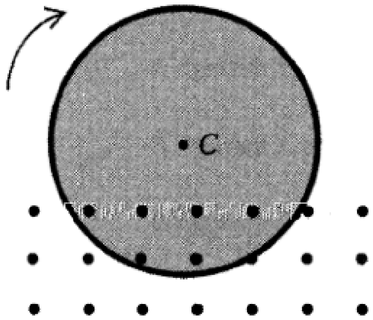
2017 Q27

27. A metal disc is rotating about its centre C with constant speed. Part of the metal disc is inside uniform magnetic field pointing into the paper as shown. An eddy current flows in the metal disc.

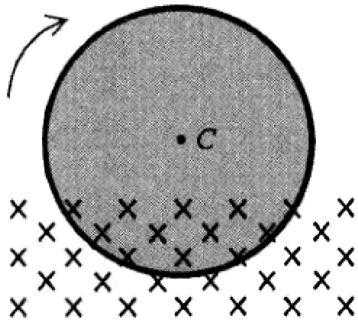


After which of the following changes will the eddy current increase ?

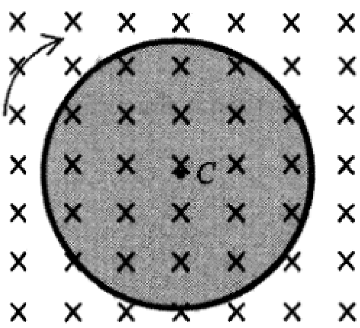
A. Reverse the direction of the magnetic field



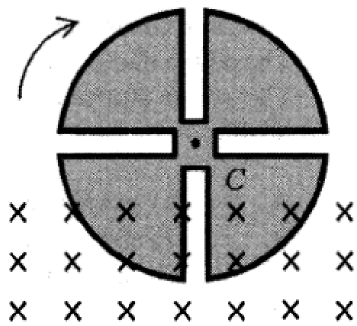
B. Increase the strength of the magnetic field



C. Apply the magnetic field over the whole metal disc

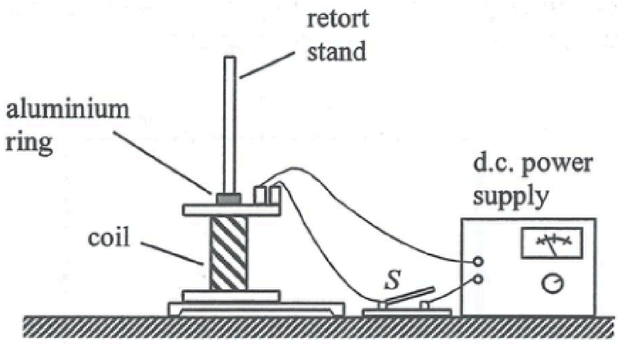


D. Cut several slits from the metal disc



2019 Q28

28.

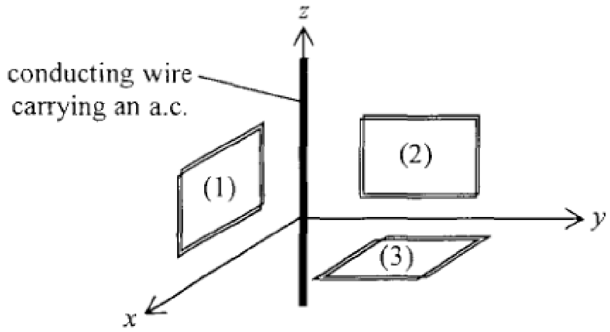


A retort stand and a coil connecting to a d.c. power supply are arranged as shown. An aluminium ring threaded through the stand is placed on top of the coil. When the switch S is closed, the aluminium ring jumps up momentarily and then falls back down. Which of the following modifications would enable the ring to rise up and float in the air ?

A. using a ring made from a lighter material
B. using a ring made from a metal of smaller resistivity
C. using a coil with the number of turns doubled
D. using an a.c. power supply instead of a d.c. power supply

2021 Q29

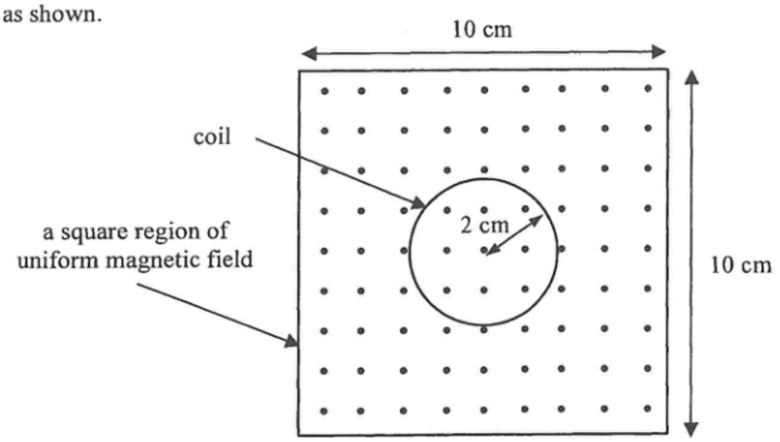
29. The figure shows three mutually perpendicular coils (1), (2) and (3) placed near a conducting wire carrying an a.c. along the z-axis direction. In which coil(s) would e.m.f. be induced ?



A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

2024 Q27

*27. A coil of radius 2 cm is placed inside a square region of uniform magnetic field pointing out of the paper as shown.



The magnetic field decreases uniformly from a strength of 5.0 T to 3.0 T in 0.8 s. What is the magnitude of the e.m.f. induced in the coil ?

A. 2.5×10^{-2} V
B. 3.1×10^{-2} V
C. 3.1×10^{-3} V
D. 6.2×10^{-3} V

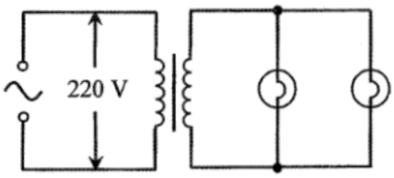
2017 Q30

*30. The input terminal of a transformer is connected to the 220 V mains supply. Ten identical light bulbs are connected in parallel to the output terminal of the transformer. All the light bulbs are working at their rated values of '3 V, 1.5 W'. If the efficiency of the transformer is 70%, what is the current drawn from the mains supply ?

A. 0.007 A
B. 0.048 A
C. 0.068 A
D. 0.097 A

2016 Q31

*31.



In the above circuit, each light bulb works at its rated value '22 W, 11 V'. The current in the primary coil is 0.25 A. Find the efficiency of the transformer.

A. 20%
B. 40%
C. 64%
D. 80%

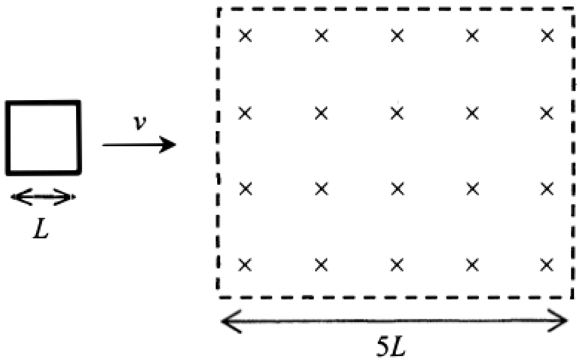
2025 Q29

*29. A student uses a search coil to measure the magnetic field of a permanent magnet. However, no reading is observed on the CRO connected to the search coil. Which of the following is the possible reason ?

A. The magnet is not strong enough.
B. The magnetic field produced by the magnet is constant.
C. The sensitivity of the CRO is too low.
D. The size of the search coil is not large enough to pick up sufficient magnetic flux.

2012 Q31

31.



A square metal frame of length of side L moving with constant velocity v passes through a region of uniform magnetic field of width $5L$ as shown. What is the total time period during which a current is induced in the frame ?

A. $\frac{L}{v}$

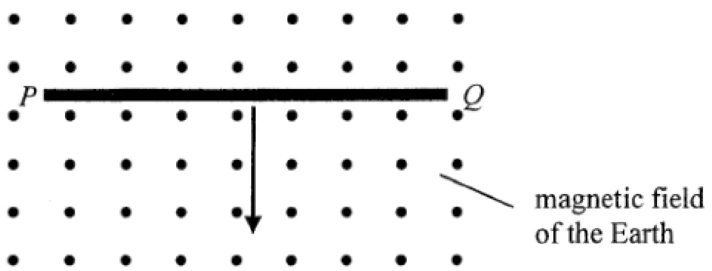
B. $\frac{2L}{v}$

C. $\frac{3L}{v}$

D. $\frac{4L}{v}$

2012 Q32

32. A copper rod PQ is placed horizontally as shown below. It is released and then falls vertically, cutting across the magnetic field of the Earth pointing out of the paper. Neglect air resistance. Which of the following statements is/are correct ?

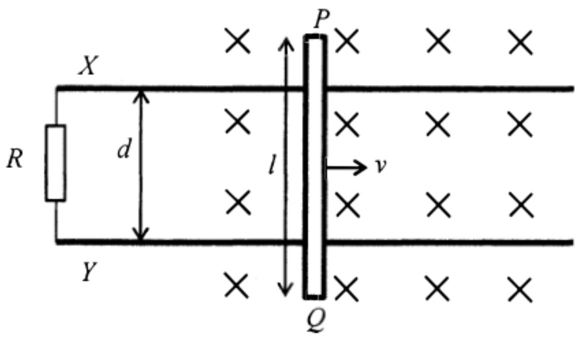


(1) A voltage is induced across PQ.
(2) A steady induced current is generated in the rod.
(3) Due to the effect of the Earth's magnetic field, the copper rod falls with an acceleration less than the acceleration due to gravity.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

2017 Q28

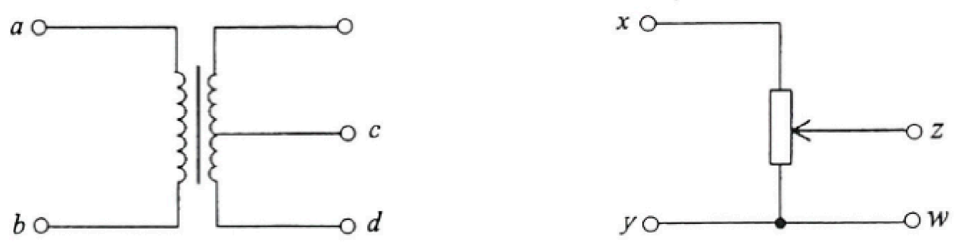
*28. A metal rod PQ of length l is moving along smooth horizontal metal rails X and Y with constant speed v in a uniform magnetic field of magnetic field strength B pointing into the paper. The metal rails X and Y are separated by a distance of d and are connected to a resistor of resistance R as shown.



Which of the following descriptions about the induced current is correct ?

	magnitude	**direction**
A.	$\frac{Blv}{R}$	from X to Y through R
B.	$\frac{Blv}{R}$	from Y to X through R
C.	$\frac{Bdv}{R}$	from X to Y through R
D.	$\frac{Bdv}{R}$	from Y to X through R

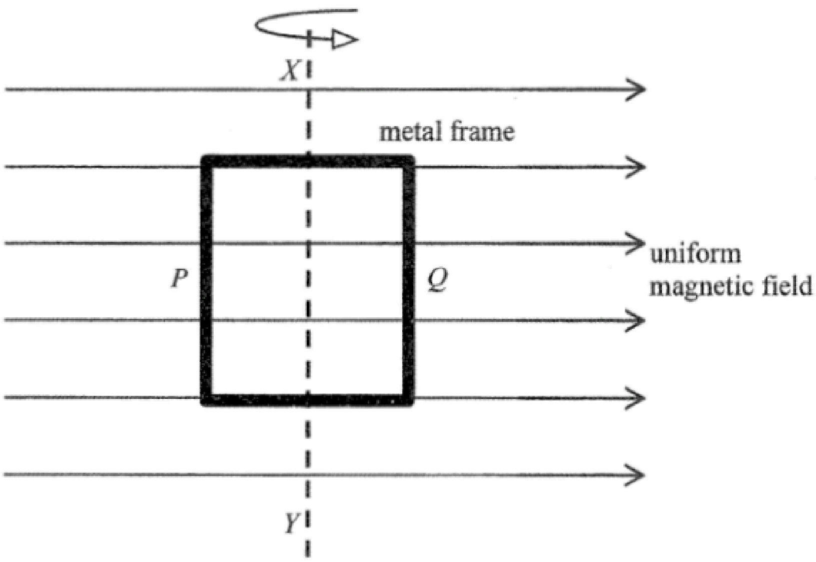
*30. In the circuits below, if a 12 V sinusoidal a.c. is applied across *ab* and across *xy* respectively, the voltages across *cd* and *zw* are both 6 V. Now if a 6 V sinusoidal a.c. is applied across *cd* and across *zw* respectively, what would be the voltages across *ab* and *xy* respectively ?



	voltage across *ab*	**voltage across *xy***
A.	12 V	12V
B.	12 V	6 V
C.	6 V	6 V
D.	12 V	0 V

2015 Q24

24.

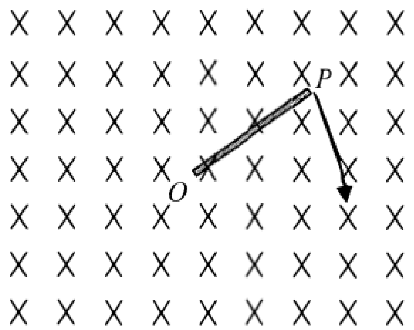


A rectangular metal frame is made to rotate steadily about its axis XY in a uniform magnetic field. At the instant shown, the frame is in the plane of the paper and side P is moving out of the paper while side Q is moving into the paper. Which statement is **INCORRECT at this instant** ?

A. The induced e.m.f. in the frame is at a maximum.
B. The induced current produced in the frame is flowing in anti-clockwise direction.
C. The magnetic force acting on side P is in a direction pointing into the paper.
D. The magnetic forces acting on the frame produce a moment opposing the frame's rotation.

2013 Q29

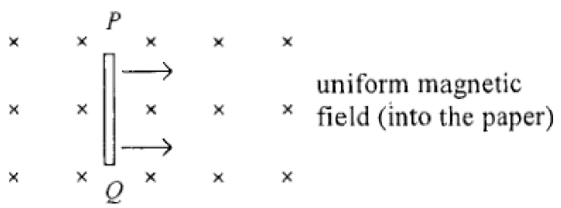
29.



A metal rod OP is rotated about O in a clockwise direction in the plane of the paper with a uniform magnetic field pointing into the paper. Which statement is correct ?

A. An induced current flows in the rod from O to P.
B. An induced current flows in the rod from P to O.
C. E.m.f. is induced in the rod with end O at a higher electric potential.
D. E.m.f. is induced in the rod with end P at a higher electric potential.

26. When a copper rod PQ moves with a constant velocity across a uniform magnetic field as shown, an e.m.f. is induced across the rod.



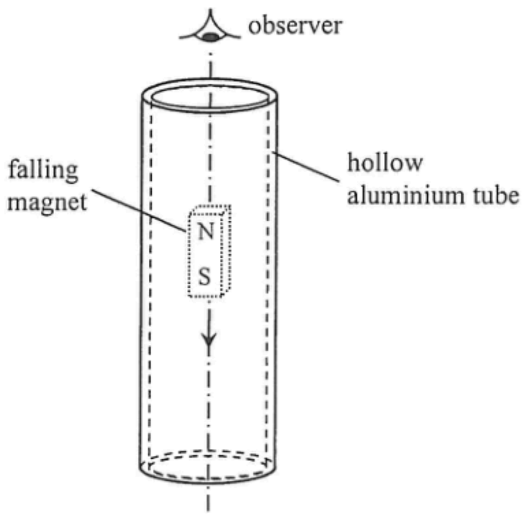
Which of the following statements is/are correct ?

(1) The magnitude of the induced e.m.f. depends on the length of the rod.
(2) Rod PQ acts like a cell providing an e.m.f. with P being its positive terminal.
(3) There is a force acting on the rod to oppose its motion.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

2020 Q28

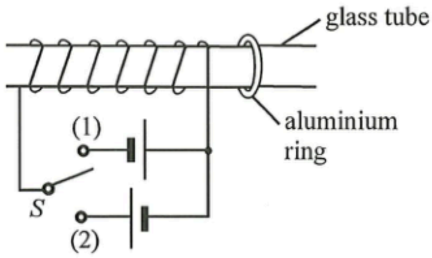
28.



When a small strong magnet falls through a hollow aluminium tube as shown, eddy currents are induced. Which of the following correctly describes the direction of current induced in the tube when viewed by an observer from above ?

A. clockwise both above and below the magnet
B. anti-clockwise both above and below the magnet
C. clockwise above the magnet and anti-clockwise below the magnet
D. anti-clockwise above the magnet and clockwise below the magnet

27. A solenoid is tightly wound around a smooth horizontal glass tube as shown. A movable aluminium ring is threaded on the right side of the tube.

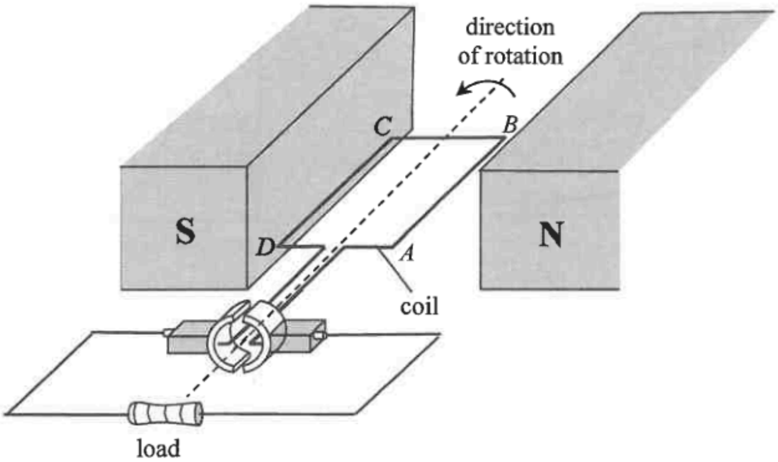


Initially the ring is at rest and the 2-way switch S is in open circuit. In what direction would the ring move at the moment when S is connected in turns to (1) and (2) ?

	when S is connected to (1)	**when S is connected to (2)**
A.	to the left	to the left
B.	to the right	to the right
C.	to the right	to the left
D.	to the left	to the right

2022 Q28

28.



The figure shows the structure of a simple generator. Which of the following statements is/are correct ?

(1) The magnetic force acting on side *AB* of the coil is upward at the instant shown.
(2) The commutator would reverse the direction of current in the coil whenever the coil passes its vertical position.
(3) The current flowing through the load is an unsteady d.c.

A. (1) only
B. (2) only
C. (3) only
D. (1) and (3) only

2014 Q27

27.

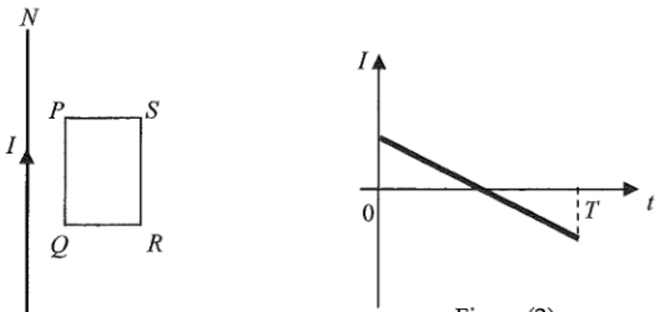


Figure (1) Figure (2)

A long straight current-carrying wire MN and a rectangular coil $PQRS$ are fixed in the same plane as shown in Figure (1). The current I is taken as positive when it flows from M to N and it varies with time t as shown in Figure (2). The direction of the induced current in the coil during the time interval $0 - T$ is

A. first anti-clockwise and then clockwise.
B. first clockwise and then anti-clockwise.
C. anti-clockwise throughout.
D. clockwise throughout.

2016 Q29

*29. A student uses a search coil to study the strength of the magnetic field inside a long solenoid which is connected to an a.c. signal generator set at a certain frequency. Which of the following can improve the accuracy of this experiment ?

(1) Ensure the plane of the search coil is perpendicular to the field lines.
(2) Increase the signal generator's frequency and use the same current as before.
(3) Set the axis of the solenoid along an east-west direction to avoid the effects of the Earth's magnetic field.

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

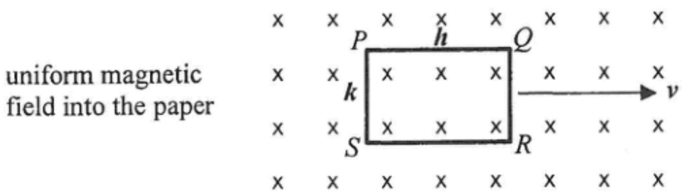
2026 Q28

*28. The input terminals of a transformer are connected to a 220 V mains supply. Its output power is 18 W and its output voltage is 5.0 V. If the efficiency of the transformer is 85%, what is the current drawn from the mains supply ?

A. 0.070 A
B. 0.082 A
C. 0.096 A
D. 0.55 A

2026 Q30

*30. A rectangular copper loop *PQRS*, with its plane perpendicular to a uniform magnetic field of flux density $\boldsymbol{B}$, is moving with velocity v in a direction perpendicular to the field as shown. The length and width of the loop are $\boldsymbol{h}$ and $\boldsymbol{k}$ respectively.



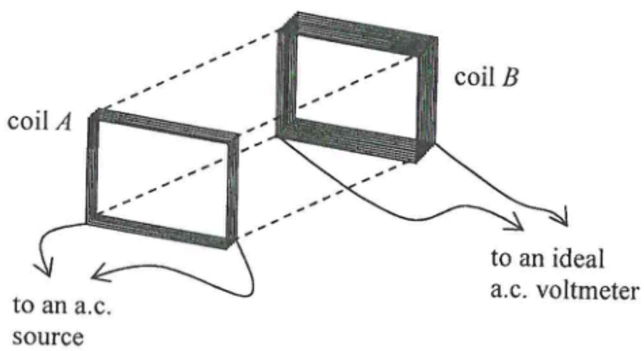
Which of the following statements is/are correct ?

(1) The potential difference across *PQ* is $\boldsymbol{Bhv}$.
(2) The potential difference across *QR* is $\boldsymbol{Bkv}$.
(3) There is **no** induced current flowing through the loop.

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

2026 Q31

*31. Coil *A* and coil *B* are placed close and facing each other with the number of turns of coil *B* being doubled that of coil *A*. Coil *A* is connected to an a.c. source while coil *B* is connected to an ideal a.c. voltmeter as shown. The voltmeter shows a reading after the a.c. source is turned on.



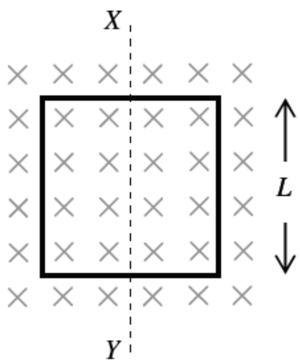
Which of the following will increase the voltmeter reading ?

(1) The voltage of the a.c. source is increased.
(2) A thicker wire is used in coil *B*.
(3) A thin and large plastic sheet is placed between the two coils and parallel to them.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

pp Q33

*33.

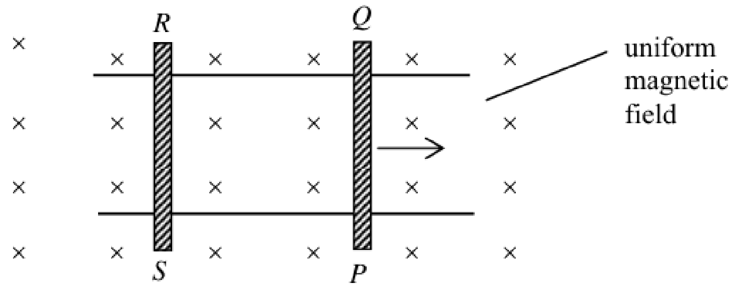


A square metal frame of side length L is placed inside a uniform magnetic field B as shown. What is the change in magnetic flux through the frame when it is rotated about the axis XY by 90° and 180° respectively ?

	90°	**180°**
A.	0	0
B.	0	$2BL^2$
C.	BL^2	0
D.	BL^2	$2BL^2$

sap Q31

31.



The figure shows conducting rods PQ and RS placed on two smooth, parallel, horizontal conducting rails. A uniform magnetic field is directed into the plane of the paper. PQ is given an initial velocity to the right and left to roll. Which statement is **INCORRECT** ?

A. The induced current is in the direction $PQRS$.
B. The magnetic force acting on rod PQ is towards the left.
C. Rod RS starts moving towards the right.
D. Rod PQ moves with a uniform speed.